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Started	Tuesday, 20 October 2026, 9:14 PM
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Duration	4 mins 33 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the list contains duplicate elements
- c. When the list is sorted
- d. When the search algorithm finishes

Question **2**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. At the beginning of the list
- c. In the middle of the list
- d. In the right sublist ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It is slower than linear search for small lists
- c. It requires the list to be sorted ✓
- d. It can only be applied to large lists

Question **4**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99

b. 90 and 100

c. 89 and 94

d. 90 and 99 ✓

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The list is sorted

c. The search continues in the right sublist

d. The search ends successfully ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. The list must be sorted

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly

b. Checking each element sequentially ✓

c. Sorting the list before searching

d. Skipping every second element

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is very small
c. When the list is sorted ✓
d. When the list contains duplicate elements

Question **9**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓
b. Both (1) & (3)
c. Binary search
d. Hash search

Question **10**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Linear search ✓
b. Interpolation search
c. Hash search
d. Binary search

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search is more efficient than binary search on large lists.
c. Linear search requires the list to be sorted.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(log n)
- c. O(n log n)
- d. O(1)

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(1)
- c. O(log n) ✓
- d. O(n log n)

Question **14**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search does not require dividing the list
- b. Binary search can find multiple instances of the target element
- c. Binary search works on unsorted lists
- d. Binary search has a lower time complexity on large, sorted lists ✓

Question **15**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✗
- b. The higher middle element is chosen as the middle element
- c. The search stops
- d. The middle element is chosen randomly